What a Recipient Can Recover from a Useful Prediction: An Evaluation of Feature Defenses and Output Contracts on Stored Representation Models

Anonymous authors

Abstract

Methods that protect a declared attribute in a learned representation come with checks that cover a particular object and a particular class of attackers. A recipient served by a multi-purpose model may receive more than that object: a task output instead of, or together with, protected features, and two recipients that serve the same person may link what they receive. We ask what a recipient can recover about the attribute from everything it receives, alone or linked with a second recipient, and how useful the release remains. On stored Adult and HMDA representation models we apply each method's own check and then attack every released surface with one slate of attackers fitted on labelled releases of other records. Official LEACE meets its linear bound, yet nonlinear attackers recover the attribute about as well as from untreated features (AUC 0.817 and 0.863, against 0.815 and 0.870). The guarantee covers linear classifiers, so this recovery lies outside its scope and does not contradict it. Releasing the untreated model's task output beside noisy or FARE features restores recovery to 0.76–0.79. When each treatment releases a useful logistic-regression head, FARE lowers sex recovery from the head's score from 0.773 to 0.539, but a same-budget tree without the fairness term accounts for about half of that reduction, and FARE's accuracy is not shown to stay within one point. In the one coalition we evaluated, two recipients granted perfect linkage of their outputs recover more than either alone, even from decisions. All results are development evidence on previously used data under a declared threat model; we propose no new method and no privacy guarantee.

1 Introduction

Methods that protect a declared attribute in a learned representation are stated and checked on that representation. LEACE erases the linearly available information about the attribute: on the distribution it is fitted to, no linear classifier reading the erased representation does better than a constant predictor under any convex loss [1]. FARE maps records to a finite set of cells and certifies, with high probability, a bound on the demographic-parity distance of any classifier that reads only the cell [12]. The purpose-specific encoders we study were trained so that a linear probe's R^2 for each disallowed attribute stays below a threshold. Each of these promises is precise about two things, the object it covers and the attackers it covers, and neither has to coincide with what a recipient of the system holds.

A release policy is defined by what each recipient receives. As an illustration, and not a description of any documented deployment, consider a service that holds records about people and serves two recipients with different permitted purposes: one receives a prediction of income, the other a prediction of an occupation group, and neither may use a declared attribute such as sex or race. For each person it serves, a recipient receives the output of its own task (a score, a probability vector or a decision) and possibly the features that produced it. The policy is written per recipient, but the person is shared. A recipient may know whom an output describes, so two recipients that serve the same person could link their outputs by record, and a recipient that knows the protected attribute for some of the people it serves could learn how outputs relate to that attribute. Under these conditions a defense applied to the features addresses only part of what recipients see: the task output may carry what was removed from the features, a recipient may hold features and output together, and two recipients may pool linked outputs. We know of no source that documents a deployment with all of these properties, so we do not claim that one exists.

These conditions raise a narrow question, which this paper studies. For stored representation models and published defenses, what can a recipient recover about a declared protected attribute from everything it receives, alone or linked with one other recipient, and how useful is that release for the recipient's task? We answer it under a declared threat model (Section 3). The recipient holds exactly its release, the attacker learns from a labelled sample of other records' releases from the same population, and a pair of recipients is granted perfect linkage. Other access, such as a recipient that also holds a person's inputs, public model parameters or permission to query repeatedly, defines a different threat model, which we leave untested rather than dismiss.

We study six stored purpose-specific encoders trained on Adult and HMDA. For each, we first apply each method's own protection check and then measure recovery from every surface a recipient could receive: the representation, the task output in four formats that carry different amounts of information, features and output together, and the outputs of two recipients linked by record. The same slate of linear, tree and neural attackers, fitted and selected on held-out roles, attacks every surface, and each recovery figure is read beside the held-out utility of the output that is actually released. A tree of the same budget as FARE's but without its fairness term separates the effect of that term from the effect of compression.

The results separate three situations that a single recovery figure would conflate. A method can pass its own check while recovery continues outside the check's scope: official LEACE meets its linear bound on every map, and nonlinear attackers recover the attribute about as well as from untreated features. The encoders' own constraint is a different case, because their stored linear check fails on 11 of 42 combinations of purpose–attribute pair and encoder seed, which is a failure of the method's own test. Finally, recovery can come through a surface that a feature defense does not cover: releasing the untreated model's output beside defended features restores recovery, and output-only recovery exceeds what the true label alone allows. Utility changes how low recovery should be read. The frozen task heads' hard decisions leak little, but they barely beat a constant. When every treatment releases a useful head, FARE's score recovers much less than the untreated or LEACE score, yet about half of that reduction is reproduced by a same-budget tree without the fairness term, and FARE's accuracy is not shown to stay within one point. In the one coalition we evaluated, two recipients with perfectly linked outputs recover more than either alone, even when each receives only decisions.

None of this shows that the evaluated methods are broken; where their promises could be checked, they held. The results show what a method has to address if its aim is to limit what recipients can learn: the complete release under a stated access model, the utility of the output that is actually released, a capacity-matched control for any effect claimed for a fairness term, and the linked view of recipients that serve the same people (Section 7). Our contribution is this evaluation and its measurements, with their qualifications. The individual observations, that linear protection does not stop nonlinear recovery, that outputs leak beyond labels, that hard labels are not a defense in general and that combined releases can leak more, are known (Section 4); we measure them together, with utility, on the same models and under one stated threat model. Every assessment row had been used by earlier analyses of these models, and registering endpoints before fitting does not turn these rows into fresh confirmation. Intervals are conditional on the fitted heads and attackers. We propose no new algorithm, no population privacy guarantee and no composition theorem, and we make no use of an earlier, retired bound from linear R^2 to classifier accuracy.

2 Setting

Stored models. We use six stored purpose-specific encoders from earlier work, trained on Adult and HMDA with encoder seeds 0, 1 and 2. Each encoder passes a record through a frozen, seeded, randomly initialised backbone, which is not pretrained, and a trained adapter per purpose. It produces a 64-dimensional representation for each purpose and carries a frozen task head for each purpose. The encoders were trained with a linear constraint on declared disallowed attributes, a one-hot ridge R^2 computed on each training minibatch and held at or below 0.05; their stored check is a different object, an in-sample one-hot ridge R^2 on the test split with the same threshold. The primary cells are Adult `income_prediction` with protected attribute sex and HMDA `underwriting` with race; secondary analyses use all 14 stored purpose–attribute pairs.

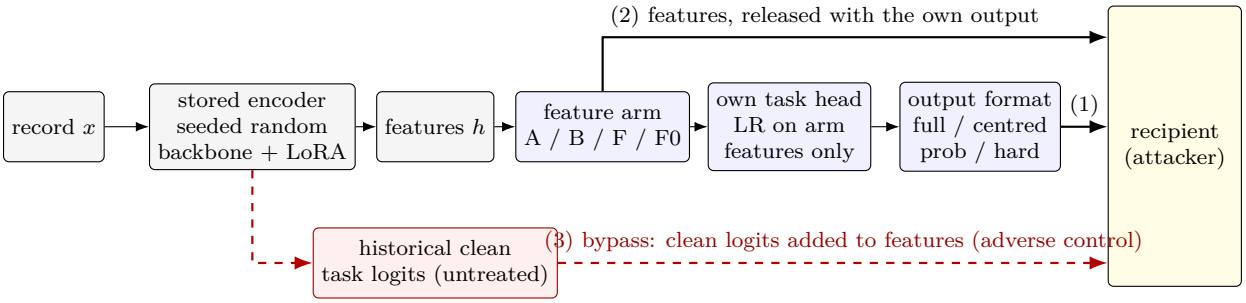


Figure 1: Recipient view. Solid paths are permitted releases: (1) output only, in one of four formats; (2) features of a treatment (A untreated, B LEACE, F FARE, F0 zero-fairness FARE tree) released with the output of a task head fitted only on those features. Path (1) is drawn for the head fitted on a treatment's features (Experiment 4); in Experiments 1–3 the released output is that of the stored frozen head, computed from untreated features. The dashed path (3) adds the historical clean logits to the features; it is the bypass, evaluated only as an adverse control.

Recipients and release contracts. Figure 1 shows the paths. Besides features alone, which we report for interpretation, a recipient receives (1) a task output only; (2) features together with the output of a head fitted only on those features; or, as an adverse control, (3) features together with the historical clean task logits of the untreated model. The third path is a bypass that a release policy should not permit. Outputs are released in one of four formats. For a K-class logit vector l, write the common offset $c = \frac{1}{K}\sum_k l_k$ and the centred logits $l - c\mathbf{1}$; for $K = 2$ the centred logits are determined by the margin $d = l_1 - l_0$. Probabilities and decisions do not depend on c (softmax shift invariance [7, §4.1]). The formats are *full* logits, *centred* logits, *probabilities*, and the *hard* decision.

Feature treatments. A: the untreated representation. B: official target LEACE [1] (concept-erasure 0.2.4), fitted on the owner's `defense_fit` role. F: official FARE [12], whose release is the one-hot tree cell. Its configuration is chosen from a frozen grid among those whose probe accuracy on `attacker_val` is within one point of the untreated probe's, as the one with the lowest validation recovery for our nonlinear attacker. F0: the same FARE configuration with fairness weight zero, that is, a tree of the same budget, grown on the task label without the fairness term, so that it only compresses. Experiment 1 also adds Gaussian noise to the representation at the smallest scale σ^* whose validation recovery for our nonlinear attacker is at most 0.55 ($\sigma^* = 2$ on Adult and 4 on HMDA). Both defenses were therefore tuned against the evaluated attacker slate on the attackers' own selection rows, a selection that favours them when they are scored against the same slate. No defense grid was expanded and no configuration was re-selected after assessment.

3 Threat model and access

The recipient and its release. A *recipient* is a party permitted one purpose. For each record it serves, it receives that purpose's release under the contract in force and nothing else about the record: not the record's inputs, its true task label or its protected attribute, and not the model's parameters, gradients or internal states (defense maps, heads, or a representation that the contract does not release). Each record is released once, and no recipient submits inputs of its choosing. For the noise treatment, noise is drawn once per record for each of three release seeds; the three realizations are attacked separately and averaged, never pooled for one record. The label-only reference of Section 6.2 is a comparator, not a recipient: it reads the scored record's true task label.

Why this access. The claims we evaluate concern a release policy, so the adversary holds what the policy delivers. That part of the access describes the service itself: an authorized recipient necessarily possesses the information delivered to it. Two further capabilities are granted by the evaluation and are not established for any deployment. The first is linkage. Two recipients that serve the same person may link their releases by record, and the linked-recipient evaluation asks whether protection that holds for each recipient survives when they pool. We grant perfect linkage, the attacker-favourable limit: a low pair recovery cannot be blamed on

linkage error, while a high one may overstate what a coalition with imperfect linkage would recover. We do not assume that contracts or governance prevent linkage, and we do not claim that pooling is unavoidable. The second capability is labelled auxiliary data: releases of other records from the same population together with their protected attribute. This models a recipient that knows the attribute for some of the people it serves and can therefore learn how outputs relate to the attribute, as in attribute-inference work in which attackers train on labelled examples or on people who disclosed the attribute [11, 21]. The availability of such data, its size and its match to the scored population are assumptions of the benchmark; we do not show that a recipient could obtain such data in practice. A stronger attacker could recover more, but that alone does not justify granting more access. Each further capability defines a separate threat model that needs its own justification.

Attackers. Our attackers are supervised. Each is fitted on the releases of the `attacker_fit` records labelled with the protected attribute, chosen by log loss on the labelled `attacker_val` records, and scored on `assessment` records that no attacker was fitted or selected on (Section 5). Tree and neural attackers also hold out an internal 10% of `attacker_fit` for early stopping. The sizes of the two labelled roles fix the attacker's data budget; they differ between experiments and are listed in Table 1. The labelled sample is a requirement of these supervised attackers, not of every conceivable attack, and attacks that need no labelled sample, for example ones built from a known population law, are not evaluated. Attackers are refitted to the releases of the defended system and so adapt to the distribution that the defense emits; only in that limited sense are they defense-aware. Knowing a defense's published algorithm and objective is different from holding its fitted parameters. A recipient may know the former, but the attackers whose recovery we report use neither beyond refitting to the defended releases. Records are assigned to roles by a hash of a canonical record key, which also groups duplicate records; keys and row identifiers align releases and are never attacker inputs.

What recovery measures. Recovery is the held-out AUC that our attackers achieve. It is a measured property of a fixed attacker slate under this threat model, not a privacy guarantee. A difference in recovery between two treatments can reflect information about the attribute that one treatment removed, limits of our attackers at this data budget, or both, and our measurements do not separate the two. A deterministic or real-valued release need not be reversible: a function of the representation can merge records, and a deterministic map to probability vectors, which stays within the probability-output contract, can remove information about the attribute. Conversely, we do not establish that any evaluated release preserves the population distribution of the attribute given the record. A high recovery therefore does not show that a release kept all of the attribute's information, and a low recovery does not show that the information is gone.

Access in each experiment. Table 1 gives, for each experiment, what the recipient receives, the labelled sample, how the attackers are fitted, selected and scored, what they know about the defense, how releases are linked, and what remains untested. The attackers of each experiment are described as they were implemented; no experiment is credited with a capability that only a later evaluation added. One historical attacker exceeds the access described above. On the noise treatment only, Experiment 1 also ran two exploratory Gaussian likelihood-ratio attackers that know the noise scale; one of them is fitted on clean, pre-noise representations of the auxiliary records, which no contract releases. They are not insider, repeated-query or white-box attacks: each sees one noisy release per record and none uses model parameters or the target's clean representation. At σ^* they recovered 0.546 and 0.563 on Adult and 0.526 and 0.575 on HMDA, against 0.528 and 0.513 for the slate (exploratory values with unadjusted intervals and no paired test). The noise rows of Table 2 are therefore not the strongest attack that was run on that surface.

Threat models outside this evaluation. The following are distinct threat models that we do not test. Several involve no breach of the provider: a recipient may legitimately hold a person's inputs, a model may be published, and a service may permit repeated queries. Some are governed by release-policy choices such as rate limits, one release per person, query auditing and whether to publish the model; others need security controls. We claim no protection against any of them and do not dismiss them: recipients that also hold a record's inputs or later learn its true label; public or leaked model parameters or gradients; permitted repeated, adaptive or chosen-input queries, including releases from several model versions; attacks that

Table 1: Access in each experiment. Common to every attack: one release per record (noise: one draw per record for each of three release seeds, attacked separately); attackers fitted on labelled `attacker_fit` releases, selected by `attacker_val` log loss and scored on `assessment`; no attacker receives a record's inputs, model parameters, gradients or defense maps, and record keys are never attacker inputs. Labelled sample: rows of `attacker_fit` / `attacker_val`. The label-only reference reads the scored record's true task label and is a comparator, not a recipient.

Experiment	What the recipient receives	Labelled sample (Adult; HMDA)	Defense knowledge, linkage, untested access
Encoders' stored check (reproduced in Exp. 1)	a check, not an attack: ridge fitted and scored in sample on the 64-dimensional representation of every test record	none; uses the scored records' own labels (15,060; 13,661)	the method's own test; covers only linear prediction from the representation
1. Feature removal (all 14 pairs)	representation (untreated, LEACE, noise at σ^*); representation with the untreated head's full logits; full logits only	7,571 / 2,239; 6,794 / 2,089	slate refitted to releases; σ^* chosen on `attacker_val` with the slate; exploratory likelihood-ratio attackers know σ (see text). Features and output held by one recipient. Untested: repeated or fresh-noise queries, own-head outputs, other formats, pairs
2. Finite releases and output formats	FARE or zero-fairness cell (one-hot) or representation; either with the untreated head's full logits; frozen-head output as full logits, probabilities or hard decision	6,065 / 2,235; 5,388 / 2,082 (after 1,500; 1,385 rows for FARE certification)	slate refitted to releases; FARE configuration chosen on `attacker_val` with the slate; a cell-conditional attacker on some finite releases (descriptive). No pairs. Untested: attacks using FARE's tree or objective, noise-aware attacks, repeated queries
3. Output diagnosis	frozen-head output in six formats (full, centred, probabilities, offset, centred with offset, hard) and two format banks; Adult income and occupation-group outputs concatenated for one linked pair; features, own output or both on an easier task	as Exp. 2 (pair and easier task: Adult only)	no defense on the frozen and pair outputs; FARE configuration on the easier task chosen on `attacker_val` with the slate. Pair: perfect linkage granted by aligned rows, linked labelled sample, single-recipient attackers among the candidates. Untested: imperfect linkage, more than two recipients, pairs that include features or defended outputs, HMDA pairs
4. Useful heads (Adult income / sex)	output of a logistic-regression head fitted on each treatment's features, in four formats and two banks; features with own output; features only; features with the untreated head's logits	as Exp. 2 (Adult)	slate refitted to releases; F is Exp. 2's configuration. No pairs. Untested: repeated queries, other head families, HMDA

exploit a defense's published objective or its fitted parameters beyond refitting; insiders; imperfect linkage; coalitions of more than two recipients; and auxiliary samples larger or smaller than ours, drawn from another population, or containing the auxiliary records' inputs or task labels.

4 Related work

Concept erasure and its evaluation. Linear concept erasure aims to stop linear classifiers from recovering a concept: INLP [17] by iterative nullspace projection, R-LACE [18] through a linear minimax game, and LEACE [1], which guarantees it in closed form, for the moments of the data it is fitted on, with the least squared change among linear erasers. Such guarantees do not extend to nonlinear recipients: INLP and R-LACE report nonlinear recovery after linear erasure, kernelized erasure does not transfer across kernels [19], and LEACE restricts its claim to linear adversaries. Adversarially trained representations show the same pattern: post-hoc attackers recover what the in-training adversary could not [3] and still read the attribute from adversarially censored representations [21], and independent post-hoc adversaries are a standard check [16]. Our LEACE measurements are a further instance, not a refutation of LEACE's theorem.

Leakage through task outputs. Model-inversion and attribute-inference attacks exploit confidence scores [4, 5]; label-only attacks can match confidence-based ones [15], and masking confidences does not stop label-only membership inference [2]. Wang et al. [25] compare leakage from predictions with that implied by the true labels at matched task performance; we report a label-only reference in the same spirit. A multiclass log-linear head on a linearly guarded representation can still reveal the erased concept, although a binary one cannot [20]. Black-box attribute inference on training records rarely exceeds what an adversary with the same distributional knowledge achieves by imputation [10], so we report constant and label-only references. Coarsening outputs, for example rounding confidences [4], is an old countermeasure; releasing probabilities rather than logits leaves calibration [8] and every proper task loss unchanged.

Fair representations and trade-offs. When the label depends on the protected attribute, fair representations trade accuracy for protection [26, 27]. Stadler et al. [22] prove that, for finite discrete data with full support and a strictly positive label posterior, a representation that carries more information about the task label than a level γ cannot keep, for *every* attribute, the inference gain of an adversary who sees it together with the label, over one who sees the label alone, within that level. The violating attribute is constructed, and the authors note that leakage of a single attribute can sometimes be restricted. We report an analogue of their label-only adversary as a descriptive comparison, not a floor: in our setting the releases are computed from the inputs alone, so they can reveal less, or more, than the true label does. In Experiment 2 the frozen heads' full outputs revealed more than the label-only reference and their hard decisions revealed less (exploratory). Adversarial methods such as LAFTR [14] lack guarantees against unseen classifiers; FCRL [9] evaluates with a common downstream protocol. FARE [12] restricts the encoder to a finite partition and certifies, with probability at least $1 - \epsilon$, a bound on the demographic-parity distance of any classifier that reads only the cell; its authors note that restricted encoders lose information and compare fairness-unaware k-means encoders. We add a same-budget zero-fairness tree and measure attribute recovery from the complete release.

Adversary models and combined releases. Our recipient is not a new adversary. It is the auxiliary-data attacker of attribute inference, which learns from labelled examples to read a sensitive attribute from what a model exposes [21] or from a person's public data [11], applied here to released task outputs as well as representations. Combining independent releases is a classical route to breaches [6]; recipients of differently perturbed copies of the same data can pool them, and mechanisms exist under which the pool reconstructs the data, by linear least-squares estimation, no more accurately than the best single copy [13]; an attack that combines the predictions of a biased and a fair version of one model raises attribute inference on image benchmarks [24]; and collusion-aware mechanisms exist for sequential finite-alphabet releases [23]. We measure the purpose-permission version, as a property of our attackers.

Position. In a bounded reading of the closest work, we did not find one protocol that applies, on the same stored models, each method's native check, nonlinear recovery from representation and complete output, the released output's held-out utility, a capacity-matched zero-fairness control, and recipient pairs. FARE is closest on matched utility and compression, Wang et al. [25] on utility-qualified output leakage, and Tian et al. [24] on combined releases. This is a gap in a bounded search, not evidence of priority.

5 Evaluation protocol

Experiments. The evidence comes from four experiments on the same stored models. They share the attacker slate and the endpoint discipline described below, but they were run separately, with their own registered endpoints, and they differ slightly in roles and in how intervals are computed. Experiment 1 compares untreated, LEACE and noisy representations on all 14 pairs. Experiment 2 adds official FARE and the zero-fairness tree, and compares the frozen heads' full logits with their hard decisions. Experiment 3 measures the effect of withholding the logit offset, the usefulness of the frozen heads, a two-recipient pair, and FARE's compression share on an easier task. Experiment 4 replaces the frozen heads with a useful head fitted on each treatment's features. Table 1 gives each experiment's access.

Roles. Experiments 2–4 use the same record-group roles: `defense_fit` (owner side: the encoders' own training split of 24,127 Adult and 63,704 HMDA records, on which LEACE maps, FARE trees and task heads are fitted, with a 20% head holdout for head selection), `attacker_fit` and `attacker_val` (attacker fitting and selection), and `assessment` (scoring only: 5,243 Adult and 4,764 HMDA records). Records that also occur in the encoder's training split are excluded from every scored role (17 Adult and 42 HMDA test records), and a further 20% of `attacker_fit` groups is reserved for FARE certification. The owner also uses the attacker roles: the task labels of `attacker_fit` fit the common utility probe, and the protected labels of `attacker_val` choose the noise scale and FARE's configuration (Section 2). No sensitive assessment label selects a release, grouping, penalty, attacker or utility probe. Experiment 1 used the uncleaned roles (assessment 5,250 Adult and 4,778 HMDA records); the overlapping records were retained and disclosed, and its decisions were stable when they were removed.

Attackers and recovery. Every reported surface is attacked by the same slate, with the access of Section 3: logistic regression (5 regularisation values), gradient-boosted trees (20 configurations) and an MLP (18), each fitted on `attacker_fit` releases labelled with the protected attribute. The nonlinear attacker is the better of the tree and MLP families on `attacker_val` log loss, refitted at three attacker seeds. Recovery is the macro one-vs-rest AUC on assessment over supported classes, averaged over attacker seeds, release realizations (noise only) and then encoder seeds. For features plus output, candidates that ignore one part are also eligible on validation. For the recipient pair of Experiment 3, the candidates also include each single recipient's attacker, chosen on validation like any other candidate. For some finite releases (Table 1) we additionally fit a cell-conditional attacker, which estimates the attribute's distribution within each released cell; we report it descriptively, and where both were fitted it matched the nonlinear attacker. In Experiments 3 and 4 a *bank* is a set of release formats of the same output; for each encoder seed the format whose nonlinear attacker has the lowest validation log loss is scored. The full bank includes the full logits and the offset-free bank does not. The label-only and constant references, the cell-conditional attacker, the exploratory likelihood-ratio attackers of Section 3 and the encoders' in-sample check are not part of the slate.

Utility. In Experiments 3 and 4 we report the accuracy, balanced accuracy, minority recall, log loss and gain over the constant (majority class on `attacker_fit`) of the *deployed* head whose output is released. Experiments 1 and 2 report the accuracy of a common refitted probe, and Experiment 2 also the accuracy and log loss of the head fitted on each treatment's features.

Native checks. The encoders' stored check is a historical value, the in-sample one-hot ridge R^2 on the test split with threshold 0.05, which Experiment 1 reproduced on all 42 combinations of pair and encoder seed. LEACE's implementation bound is a whitened residual cross-covariance below its tolerance on `defense_fit`. FARE's is its demographic-parity certificate ($\epsilon = 0.05$) on the certification role (1,500 Adult and 1,385 HMDA rows).

Inference. Each endpoint is a mean over encoder seeds of a per-seed statistic. In Experiments 3 and 4, standard errors come from 1,999 paired multinomial bootstrap replicates over assessment record groups, shared by every attacker fit, and intervals are simultaneous two-sided 95% Bonferroni normal intervals over the endpoint family. Experiments 1 and 2 used one-sided Bonferroni percentile bounds from 20,000 group-bootstrap replicates, and the bounds in Table 2 from those experiments are of that kind. In every experiment an endpoint passes only if its lower bound exceeds its registered target. Utility targets are a non-inferiority margin of -0.01, a gain of 0.03, and the retention contrast $\mathrm{Acc}(\text{treatment}) - 0.8\,\mathrm{Acc}(A) - 0.2\,\mathrm{Acc}(\text{const}) > 0$; the recovery contrasts of Experiments 2–4 use a target of 0.02 AUC. Experiment 1's registered endpoints compare levels with fixed bars instead (recovery with 0.55 AUC, the held-out linear statistic with 0.05 and probe accuracy with a one-point margin), and Experiment 3's frozen-head usefulness endpoints used a gain target of 0.01. An interval that excludes zero but not the target is reported as a smaller difference below the target, never as "no difference". Each experiment's protocol, code, inputs, list of fits and endpoint families were committed before its first fit, and an independent replay that does not import the analysis code recomputed every primary result. The encoders' stored check predates these experiments; Experiment 1 reproduced its values. Three changes were made after runs had begun, and each is recorded with its protocol:

Table 2: Main contrasts from Experiments 1–3 (means over three encoder seeds; the noise row also averages three release seeds). Parenthesised values are lower confidence bounds of registered contrasts: a one-sided Bonferroni percentile bound in Experiment 2 and lower ends of simultaneous two-sided Bonferroni intervals in Experiment 3 (Section 5). Several other rows are registered levels or contrasts whose bounds are reported with the experiments.

Quantity	Exp.	Adult income / sex	HMDA underwriting / race
Encoders' own linear check fails (pair×seed, all 14 pairs)	1	11 of 42	
Nonlinear recovery, untreated features	1	0.815	0.870
Nonlinear recovery, LEACE features	1	0.817	0.863
Noise at σ^*, features only / plus clean outputs	1	0.528 / 0.778	0.513 / 0.758
Full logits vs hard decision (frozen head)	2	0.776 vs 0.513: +0.263 (0.249)	0.768 vs 0.505: +0.262 (0.254)
FARE vs LEACE features	2	0.550 vs 0.812	0.501 vs 0.863
Zero-fairness tree (same budget), features	2	0.645	0.608
FARE features plus historical clean outputs	2	0.788	0.768
Offset withheld: full bank minus offset-free bank	3	+0.072 (0.062)	+0.067 (0.060)
per encoder seed	3	0.079 / 0.064 / 0.072	0.058 / −0.0003 / 0.143
Frozen head: accuracy gain over constant (target 0.01)	3	0.039 (0.031)	0.016 (0.0098), not established

a guard in Experiment 2's FARE certification code that identifies duplicate rows by record rather than by feature values, without which every certificate was unavailable; a scoring repair to the easier-task comparison of Experiment 3, found by the independent replay, which applied the registered rule for candidates that ignore one part and moved that endpoint's lower bound from 0.0001 to 0.0019 without changing its decision; and reporting repairs in Experiment 3 that changed no estimand, selection, role or decision.

6 Results

Table 2 collects the main contrasts from Experiments 1–3. Section 6.5 reports the useful-head comparison, and Sections 6.6 and 6.7 the compression share of FARE's effect and the linked recipients.

6.1 What each method's own check establishes

Three outcomes have to be kept apart: a method failing its own test, its test holding while recovery occurs outside the test's scope, and recovery through a surface that the test does not cover. The encoders' constraint is an instance of the first. Their stored linear check fails on 11 of the 42 untreated combinations (14 purpose–attribute pairs, three encoder seeds each), including HMDA underwriting/race on two of three seeds, so for those combinations the encoders do not deliver the protection that their own test describes. LEACE is an instance of the second. It meets its bound on all 60 fitted maps (including 18 fitted for a set of attributes), and its held-out squared top canonical correlation ρ_1^2 is small (point estimates 0.0006 on Adult and 0.0001 on HMDA, with unadjusted 90% upper ends of 0.013 and 0.033, on one already-used held-out split), so little linear signal is left on held-out data either. Nonlinear recovery nevertheless changes by less than 0.01 AUC (Adult +0.002, HMDA −0.007; for HMDA the unadjusted 90% interval excludes zero; Table 2). The guarantee concerns linear classifiers, and this measurement shows how much of the attribute remains available to nonlinear ones; it is not evidence against the theorem. FARE's certificate could not be assessed in the same way. At the available certification sizes it was unavailable, vacuous or weak (Adult bounds about 0.50), and its only informative bound (0.175, a secondary certificate over three race groups) belongs to a degenerate HMDA configuration with a single cell. These certificates exist only after the post-run repair of the certification code described in Section 5.

6.2 The task output as a separate surface

The third outcome concerns what a feature defense does not cover. Output-only recovery exceeds the label-only reference by 0.177 on Adult and 0.236 on HMDA, so the outputs carry information about the attribute beyond what the true label carries. When the historical clean output of the untreated model is released beside defended features, recovery returns to 0.76–0.79 for both noise and FARE (Table 2). Changing the features cannot remove what an unchanged output carries. The defenses make no claim about outputs, so this is a property of the release, not of the defenses; what it shows is that protecting the features does not limit recovery when the original output is also sent.

6.3 Low decision-level recovery and usefulness

The frozen heads' hard decisions leak little (0.513 on Adult, 0.505 on HMDA), but those decisions are weakly useful. The Adult frozen head gains 3–5 accuracy points over a constant and recalls 14–22% of high earners; HMDA underwriting usefulness is not established; and the HMDA fair_lending head predicts the same class for everyone, although its scores still vary and are informative about race and sex (centred-logit recovery 0.646 and 0.655). Low decision-level recovery here coincides with decisions that barely beat a constant. This matters for any reading of hard decisions as a defense: on these heads, the low recovery cannot be separated from the low usefulness, which is why Experiment 4 repeats the comparison with useful heads.

6.4 Withholding the logit offset

The common logit offset changes no probability, decision or task loss, yet the frozen heads release it as part of their full logits. Withholding it lowered recovery by 0.072 (Adult) and 0.067 (HMDA) AUC. The effect appeared on 5 of 6 primary encoder seeds (not on the collapsed HMDA seed 1) and on 5 of 14 stored pairs; it was not established for any of the six multiclass Adult pairs, although it passed for the five-class HMDA pricing head with race. For a refitted logistic-regression head whose logits are log-probabilities, the offset is a function of the margin, so the effect is zero by construction; that head is about twice as useful (gain 0.083), and its margin alone recovers 0.773. The result is therefore a measurement of an output contract for particular frozen heads, not a general method: it depends on how the heads were trained and disappears for heads whose logits carry no free offset.

6.5 Matched useful-head comparison

Design. On Adult income/sex, every treatment releases the output of a logistic-regression head from one common family (regularisation grid fitted on `defense_fit` minus a holdout, selected on holdout log loss) whose only runtime input is that treatment's features. The heads, LEACE maps and FARE trees are reused unchanged. Full-logit output-only, features-only, features-plus-full-output and bypass surfaces are reused, hash-verified, from Experiment 2, whose exploratory tables had already reported their levels. The centred, probability and hard surfaces and the banks are new and were fitted after the protocol was registered (3,924 model fits). Because the heads emit log-probabilities, their full logits, centred margin and probabilities are functions of one another and carry the same information (verified to 10^{-12}). The ten primary recovery endpoints use the centred and hard formats; the other nine primary endpoints concern utility.

Utility. All four heads are useful (Figure 2, Table 3): accuracy 0.832 (A), 0.831 (B), 0.822 (F) and 0.828 (F0) against a constant of 0.749, with balanced accuracy 0.72–0.73 and about half of high earners recalled. LEACE and the zero-fairness tree are non-inferior within one point. FARE is not established as non-inferior (lower bound -0.015), because seed 2 loses 2.0 points; it passes the gain and retention criteria, the latter narrowly.

Recovery from useful scores. The untreated head's score recovers sex at 0.773 and the LEACE head's at 0.759, a smaller difference of 0.015 that lies below the registered target and is carried by seed 0 alone. FARE's head recovers 0.539 and the zero-fairness tree's 0.648. Of FARE's 0.234 reduction, compression accounts for 0.125 and the fairness term for a further 0.109, on every seed. Post hoc and descriptively, a linear attacker on the untreated head's margin recovers only 0.515, while nonlinear attackers on the same scalar recover 0.773.

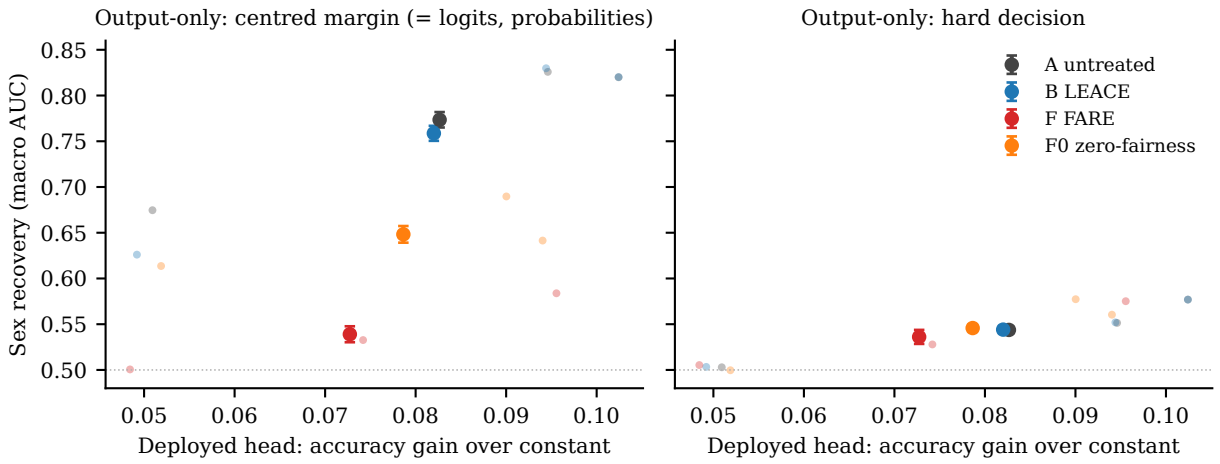


Figure 2: Useful-head comparison on Adult income/sex. Each treatment releases the output of a deployable logistic-regression head fitted on its own features. Left: the centred margin, which carries the same information as the head's logits and probabilities; right: the hard decision. Horizontal axis: the deployed head's accuracy gain over the constant; vertical axis: sex recovery (macro AUC). Large markers: means over three encoder seeds with marginal 90% intervals (the simultaneous intervals of the registered endpoints are in Table 3); small markers: individual seeds. The dotted line is chance.

This is consistent with, but does not establish, the reading that the sex signal left in the score is not linearly decodable and so lies outside a linear eraser's scope.

Recovery from useful decisions. When only the decision is released, every treatment recovers 0.536–0.546 and no contrast is established. The decision has the same accuracy as the score it comes from. Post hoc and descriptively, the untreated head's decisions leak about as much as FARE's scores (difference +0.005, 90% interval [−0.003, 0.013]) while being about one point more accurate.

Complete release. With features plus own output, the features dominate (A 0.816, B 0.817, F 0.550, F0 0.646). No offset effect was established for these heads, and none is possible in information terms, because their offset is a function of the margin. Adding the historical clean logits to F or F0 features restores recovery to 0.788 and 0.789.

Registered predictions. Four of the ten primary recovery predictions were correct. The predictions were registered after full-logit recovery of these same heads had been reported as exploratory in Experiment 2, and the protocol did not disclose that; we disclose it here. We predicted a larger LEACE effect on the score, a smaller fairness-term effect, and established hard-decision contrasts; all three were wrong.

6.6 How much of FARE's effect is compression?

The answer differs by cell. On Adult income/sex, the zero-fairness tree's features recover 0.645 against FARE's 0.550 and LEACE's 0.812 (Experiment 2). The fairness term removes 0.109 beyond compression from the head's score (Section 6.5) and 0.096 [0.076, 0.116] from the complete release of features with the head's output (a secondary endpoint of Experiment 4). On an easier task chosen by a validation-only screen (Adult occupation group with age group protected; probe accuracy 0.974 for FARE and 0.975 untreated, against a constant of 0.277), FARE's complete-release recovery is 0.151 below LEACE's, but the zero-fairness tree reproduces most of that reduction: on the same complete-release surface, FARE's additional improvement is a smaller difference of 0.0074 [0.0019, 0.0129], below the registered 0.02 target. That task is easy, but no single input column determines it. A claim about a fairness term therefore needs its capacity-matched twin.

Table 3: Primary useful-head endpoints (19; Bonferroni $z = 3.008$). Intervals are simultaneous two-sided 95% Bonferroni normal intervals; for gain and retention only their lower ends are shown. Recovery is output-only. n.e.: not established; "excl. 0": the interval excludes zero but not the target.

Endpoint (target)	Point	Interval	Decision
$\mathrm{Acc}(B) - \mathrm{Acc}(A)$ (> -0.01)	-0.0006	$[-0.0016, 0.0003]$	pass
$\mathrm{Acc}(F) - \mathrm{Acc}(A)$ (> -0.01)	-0.0099	$[-0.0152, -0.0046]$	n.e.
$\mathrm{Acc}(F_0) - \mathrm{Acc}(A)$ (> -0.01)	-0.0040	$[-0.0086, 0.0005]$	pass
Gain over constant, B / F / F0 (> 0.03)	0.082 / 0.073 / 0.079	$\geq$0.067 / 0.057 / 0.063	pass
Retention, B / F / F0 (> 0)	0.016 / 0.007 / 0.013	$\geq$0.013 / 0.001 / 0.007	pass
Centred: $\mathrm{R}(A) - \mathrm{R}(B)$ (> 0.02)	0.015	[0.011, 0.019]	n.e. (excl. 0)
Centred: $\mathrm{R}(A) - \mathrm{R}(F)$	0.234	[0.217, 0.252]	pass
Centred: $\mathrm{R}(A) - \mathrm{R}(F_0)$	0.125	[0.112, 0.139]	pass
Centred: $\mathrm{R}(B) - \mathrm{R}(F)$	0.220	[0.201, 0.238]	pass
Centred: $\mathrm{R}(F_0) - \mathrm{R}(F)$	0.109	[0.092, 0.126]	pass
Hard: all five contrasts	-0.002 to 0.010	lower bounds < 0	n.e.

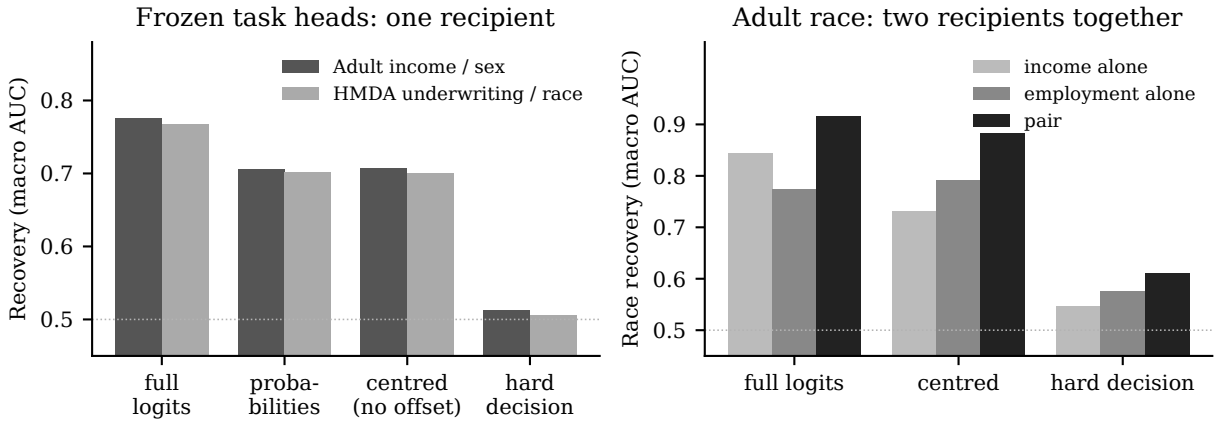


Figure 3: Left: recovery from the frozen heads' outputs by release format (Adult income/sex, HMDA underwriting/race). Right: race recovery on Adult from the income recipient, the occupation-group recipient (labelled "employment") and both linked, by format.

6.7 Linked recipients

We evaluated one coalition: two Adult recipients, each permitted only its own task output (income; occupation group) from the untreated frozen heads, whose outputs are linked by record and pooled to recover race, which is disallowed for both (Figure 3). The linkage is granted by aligning rows, and no linkage procedure is evaluated. In this family of registered secondary endpoints, the pair beats both single recipients by more than 0.02 AUC under every format: full logits 0.917 against 0.845, centred 0.883 against 0.792, and hard decisions 0.611 against 0.576 (pair minus best single +0.034, lower bound 0.023, the weakest of six passes). We report absolute pair recovery and its increment over the strongest single release separately. On seed 2 under full logits, validation selected the income-only attacker for the pair, so that seed's pair-minus-income difference is exactly zero. These are measurements with a fixed attacker slate under perfect linkage. They show that linked outputs need an evaluation of their own; they do not show that race is carried only jointly by the two outputs. AUC is not an information measure, the pair gain does not separate evidence that each output carries on its own from evidence present only in their combination, and we prove no composition or synergy result.

7 Discussion

Where the remaining recovery comes from. LEACE's promise held wherever we could check it, FARE's certificate was too weak at our certification sizes to be checked, and the encoders' own linear constraint failed its check on 11 of 42 combinations. Apart from that failure, the recovery that remained came from attacker classes or surfaces that the promises do not cover: nonlinear attackers on LEACE features, task outputs released beside defended features, and a second recipient whose outputs are linked to the first. This is why the three outcomes of Section 6.1 are reported separately. A method can meet its own promise while the release as a whole remains informative about the attribute.

What lowered recovery at matched utility, and what limits that reading. LEACE preserved the deployed head's accuracy (difference -0.0006, lower bound -0.0016; non-inferior within one point) and passed its own bound, but it barely changed what that head's score reveals (0.759 against 0.773). The zero-fairness tree preserved accuracy within one point and reduced score-level recovery to 0.648. FARE reduced it further, to 0.539, and its fairness term removed 0.109 AUC beyond the capacity-matched tree on every seed; post hoc, its score leaks about as little as the untreated decision. Three results limit what this comparison supports. FARE's accuracy is not shown to be within one point of the untreated head, so its larger reduction is not established at matched utility. On an easier task the same-budget tree reproduced most of FARE's effect, so the contribution of the fairness term depends on the cell. And each feature defense we evaluated with the unchanged output was bypassed by it. Releasing only the decision preserved accuracy by construction and gave the lowest recovery for every treatment, but it discards scores that a recipient may need. The earlier pattern of decisions leaking little was observed on decisions that barely beat a constant; with useful heads, decisions still recovered only 0.536–0.546, so weak usefulness does not explain all of it.

A measured gap (post hoc, descriptive). At matched accuracy (within one point, established), the lowest score-level recovery among the evaluated treatments is the zero-fairness tree's 0.648, while the untreated decision, which has the same accuracy and no scores, recovers 0.544. A method that released calibrated, useful scores with recovery near the decision's level would close a measured gap of about 0.10 AUC on this cell. We do not claim that such a method exists, nor that it is impossible.

What the findings ask of future methods. If the aim is to limit what recipients can learn about an attribute, the evaluation has to follow the release. A method's claim should name the surfaces it covers, including task outputs and any features released with them, and the access model under which it is evaluated, including the attacker's labelled-data budget. Recovery should be reported beside the held-out utility of the output that is actually released. An effect claimed for a fairness term should be compared with a capacity-matched control. Where recipients serve the same people, the linked view should be evaluated as well as each recipient alone. These are requirements for evaluating a method, not evidence that any particular method fails them.

8 Limitations, reproducibility, ethics and LLM assistance

Limitations. The evidence covers two datasets, a small number of primary cells and three encoder seeds, and the encoder seeds differ strongly (one HMDA encoder is heavily collapsed). All assessment rows were used before, so nothing here is fresh confirmation. Intervals are conditional on fitted heads and attackers and omit retraining variance. FARE is applied to stored representations rather than raw inputs. Rare HMDA race groups are not estimable. The threat model of Section 3 is one of several, and the threat models listed there are untested, not ruled out. Several observations in Section 6.5 are post hoc and labelled as such.

Reproducibility. For every experiment the protocol, code version, admitted-input hashes, list of fits and endpoint families were committed before fitting. Result tables, inference code and the independent replay are in the anonymous repository; per-record outputs, weights and checkpoints are kept privately with hash indexes. The encoders' public code still exposes the interface of the retired R^2-to-accuracy bound, which we do not use. The useful-head comparison used about 0.25 CPU-hours on a laptop.

Ethics. We study recovery of declared protected attributes from public benchmark data to inform release policies. Attack code is standard; we release no per-person predictions.

LLM assistance. A large language model assisted with code, analysis scripts, independent-replay scripts, literature checking and drafting. All numbers were produced by committed code. A replay that does not import the analysis code recomputed every registered endpoint; the post hoc diagnostics, the per-seed offset values and the refitted head's gain (each recomputed separately) and the intervals drawn in Figure 2 were not part of that replay. The authors are responsible for the content.

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